

Q1. What is robotics? What kind of projects can we do for a robotics competition?

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A1. Whenever we talk about robots, two questions arise: what is a robot, what does it do?

What is a robot? The word robot comes from the Czech word 'robot', which means obligatory work, servitude or forced labour. The word robot was first used in 1921 by Karl Capek in a Czech play called Rossum's Universal Robots, depicting a race of humanoid robots that turn on their masters and destroy them.

Issac Asimov, Russian science fiction writer, coined the term 'robotics' in his story 'Runaround' to denote the science devoted to the study of robots. He conceived robots as humanoids devoid of feelings, well-designed, fail-safe machines, whose brains were programmed by human beings. Anticipating the dangers and havoc such devices could cause, he postulated three rules of robotics:

1. A robot should not injure a human being or, through inaction, allow a human to be harmed
2. A robot must obey orders given by humans except when that conflicts with the first law
3. A robot must protect its own existence unless that conflicts with the first or second law

According to modern definition, a robot is a system that has:

1. One or more sensors to interact with the environment
2. A program that processes the inputs received from the environment
3. A mechanical assemblage like an arm, hand or other moving part to perform the intended function like lifting, assembling or moving something
4. An electrical assemblage that provides the motive force to the mechanical parts

What does it do? A robot is a machine designed to execute one or more tasks repeatedly, with speed and preci-

sion. As stated by the Robot Institute of America, "A robot is a programmable multi-functional manipulator designed to move material, parts, tools or specialised devices through variable programmed motions for the performance of a variety of tasks."

Robotics competitions. There is a range of robots in terms of complexity fielded in competitions. Some popular types are:

1. Obstacle-avoiding robots
2. Line-following robots
3. Wireless-controlled robots
4. Pick-and-place robots
5. Grid-following robot
6. Micro-mouse (most popular) robots
7. Soccer-playing robots
8. Wrestling robots

Q2. How does a transistor work as an amplifier?

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A2. A transistor is basically a current-control device. To understand the overall operation of the transistor as an amplifier, we will only consider the current in and out of the transistor and through the various components in the circuit. Before going into the details, let us familiarise ourselves with two terms: amplification and amplifier. Amplification is the process of increasing the strength of a signal. A signal is just a general term used for referring any particular current, voltage or power in a circuit. An amplifier is the device that provides amplification (the increase in current, voltage or the power of a signal).

A circuit of an amplifier using the npn transistor is given in the Fig. 1.

R_1 , R_2 , R_E and R_C form the base bias arrangement for a proper operating point. With T1 properly biased, direct current flows continuously, with or without an input signal, throughout the entire circuit. The direct current flowing through the circuit develops more than just the base bias; it also develops the collector voltage (V_C) as it flows through T1 and R_C . Since the current is present in the circuit without an input signal, the output signal starts at the V_C level and either increases or

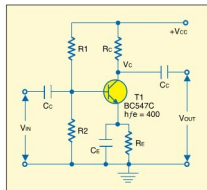


Fig. 1: Transistor amplifier

decreases. These DC voltages and currents that exist in the circuit before the application of a signal are known as 'quiescent' voltages and currents (at quiescent state of the circuit). Resistor R_C , the collector resistor, is placed in the circuit to keep the full effect of the collector supply voltage off the collector. This permits the V_C to change with an input signal, which in turn allows the transistor to amplify voltage. Without R_C in the circuit, the voltage on the collector would always be equal to V_{CC} . The coupling capacitor (C_C) is used for passing the AC input signal and blocking the DC voltage from the preceding circuit. This prevents DC in the circuitry on the left of the coupling capacitor from affecting the bias on T1. The coupling capacitor also blocks the bias of T1 from reaching the input signal source.

A signal, for example, a sine wave with a variation of a few millivolts above and below zero, is applied at the emitter—the base junction through the coupling capacitor. As the input signal goes positive, the voltage across the emitter-base junction becomes more positive. This, in effect, increases the forward bias, which causes the base current to increase at the same rate as that of the input sine wave. An increase in the current is roughly equal to h_{fe} , that is, the collector current is approximately h_{fe} (β) times the base current. With an increase in the collector current, more voltage is developed across R_C . Since the voltage across R_C and the voltage across T1 (collector to emitter) must add up to V_{CC} , an